kam: Alignment-Free Variant Detection for Duplex UMI Sequencing

v0.1 — Draft

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Abstract

Detecting somatic variants at sub-1% allele frequency in liquid biopsy samples requires accurate UMI-based error suppression and sensitive variant calling. Current pipelines chain several tools that discard molecule-level information between steps, making per-molecule strand bias and duplex evidence unavailable at the calling stage.

We present kam, an alignment-free variant detection pipeline for Twist duplex UMI panel sequencing. kam assembles duplex molecules directly from raw FASTQ, indexes k-mers with per-molecule provenance, builds a de Bruijn graph over all observed k-mers, and calls variants by scoring alternative paths against the reference path using molecule counts and strand composition. No reference alignment is performed. Called variants are left-normalised to standard VCF convention, enabling direct comparison against truth sets for SNVs, indels, and structural variants. Structural variant detection uses junction k-mer injection: breakpoint-spanning k-mers are added to the allowlist, making SV allele paths discoverable in the de Bruijn graph without specialised alignment.

We evaluate kam in two operating modes on the Twist cfDNA Pan-Cancer Reference Standard v2 titration series (375 truth variants, 24 samples, 5–30 ng input DNA, VAF 0%–2%). In discovery mode, kam detects 52–61% of truth variants at 2% VAF with precision 0.64–0.80 using 2M read pairs per sample. In tumour-informed monitoring mode (`-target-variants`), the same calls are filtered to variants matching a pre-specified target VCF: precision rises to 1.0 at all VAF levels with no change to sensitivity.

We identified and fixed two implementation bugs during evaluation: a strand bias overcounting bug that inflated the Fisher exact test sample size approximately 70-fold, and a duplex orientation comparison bug that prevented genuine duplex pairs from being grouped correctly. With both fixes applied and 2M read pairs per sample, duplex identification rates are 4.6–15.5% of molecules. With recommended filters (`-max-vaf 0.35 -min-family-size 2`) and tumour-informed mode applied, kam detects 52–61% of truth variants at 2% VAF with precision 1.0, running end-to-end in 19–35 seconds per sample on a single core. SNV sensitivity reaches 69–80% at 2% VAF; indel sensitivity is 32–39%. Sensitivity at 0.25–0.5% VAF reaches 26–46% at 15–30 ng input and 6–17% at 5 ng input.

kam is implemented in Rust and is available as an open-source workspace.

1 Introduction

Circulating tumour DNA (ctDNA) in blood plasma carries somatic variants shed from solid tumours. Detecting these variants enables non-invasive cancer monitoring: tracking treatment response, identifying resistance mutations, and detecting minimal residual disease[1]. The challenge is sensitivity. Clinically relevant ctDNA fractions range from 2% down to 0.001% of total cell-free DNA, depending

on tumour burden and stage. At these variant allele frequencies (VAFs), a single mutant molecule must be distinguished from thousands of wild-type molecules and from sequencing errors that occur at rates of 10^{-2} to 10^{-3} per base. Duplex UMI sequencing suppresses these errors by tagging each original DNA molecule with a unique molecular identifier (UMI) before amplification, then requiring that both strands of the molecule independently confirm each base call[2, 3]. This reduces the effective error rate to approximately 10^{-7} per base, making sub-0.1% VAF detection possible in principle.

Alignment-free variant detection using k-mer counting and de Bruijn graph path walking was established as a viable alternative to alignment-based calling for UMI duplex data. The standard pipeline chains HUMID for UMI-based deduplication[4], Jellyfish for k-mer counting[5], km for de Bruijn graph path walking[6], and kmtools for parallelisation and filtering. This pipeline discards molecule identity at the first step. HUMID collapses read families into single representative reads. Jellyfish then counts k-mers across all surviving reads, producing a hash table of integer counts with no record of which molecules contributed each count. km walks paths through the resulting de Bruijn graph and calls variants using a ratio threshold on these aggregate counts. The duplex strand information, the per-molecule error correction, and the family size metadata are all lost before variant calling begins.

This information loss has two consequences. First, variant evidence is inflated by PCR duplicates, increasing false positive rates at low VAF. Second, the variant caller cannot distinguish high-confidence duplex-confirmed variants from low-confidence simplex singletons. Both look the same in a table of integer k-mer counts.

We present kam, a single Rust tool that replaces the full four-tool pipeline and preserves molecule provenance from raw FASTQ reads through to variant calls. kam extracts UMIs directly from FASTQ reads, groups reads into duplex molecule families, builds consensus sequences with quality-weighted error correction, and indexes k-mers with per-k-mer `MoleculeEvidence`: the number of distinct molecules, the number with duplex support, the strand breakdown, and per-base error probabilities. Variant calling operates on molecule counts, not read counts, and uses a statistical model that weights duplex-confirmed evidence above simplex evidence.

kam supports two operating modes. In *discovery mode*, it finds all alternative paths within each 100 bp target window and reports every path that passes statistical quality filters. In *tumour-informed monitoring mode* (`-target-variants`), a VCF of expected somatic variants is provided and the caller only reports calls whose (CHROM, POS, REF, ALT) matches an entry in that set. Discovery mode is appropriate for initial somatic profiling; monitoring mode is appropriate for serial liquid biopsy where the variant panel is known from a prior biopsy.

On the Twist cfDNA Pan-Cancer Reference Standard v2 titration series with tumour-informed filtering applied, kam detects 52–61% of truth variants at 2% VAF with precision 1.0 (zero false positives), running in 19–35 seconds per sample on a single core.

Section 2 covers duplex UMI sequencing and k-mer variant detection. Section 3 reviews existing tools and their limitations. Section 4 describes kam's architecture, from molecule assembly through statistical calling. Sections 5 and 6 present benchmark results on the titration dataset.

2 Background

2.1 Duplex UMI sequencing

Standard next-generation sequencing produces per-base error rates of 10^{-2} to 10^{-3}, set by the accuracy of the sequencing chemistry. Detecting a variant at 0.1% VAF requires distinguishing a true signal from errors that occur at 10 to 100 times the signal frequency. Duplex UMI sequencing solves this through two layers of error correction[2].

Before PCR amplification, each original DNA molecule receives a pair of UMI tags, one ligated to each strand. All PCR copies of that molecule carry the same UMI pair. After sequencing, reads sharing a UMI pair are grouped into a *molecule family*. The first error correction step collapses all reads from one strand of a family into a *single-strand consensus sequence* (SSC) by majority vote weighted by base quality. Positions where most reads agree are called with high confidence; positions where reads disagree are flagged or masked. This reduces the error rate to approximately 10^{-3} to 10^{-4}, limited by systematic errors that affect all copies of one strand identically.

The second step compares the forward-strand SSC to the reverse-strand SSC. A sequencing or PCR error on one strand is unlikely to produce the identical error on the complementary strand independently. Positions where both SSCs agree form the *duplex consensus sequence* (DCS), with an effective error rate of approximately 10^{-7}: the product of the two independent per-strand error rates[3]. This three-order-of-magnitude improvement over single-strand consensus is what makes sub-0.1% VAF detection feasible.

The Twist Biosciences UMI adapter system[7] uses the read structure `5M2S+T` on both R1 and R2: 5 bases of random UMI, 2 bases of skip sequence (a fixed monotemplate spacer from the ligation chemistry), and the remaining template sequence. The 5-base UMI provides $4^5 = 1{,}024$ possible UMI values. Duplex strand pairing uses a canonical form: given UMI pair (A, B) from R1 and R2, the canonical identifier is $\min(A{+}B,\ B{+}A)$ by lexicographic order. Reads with UMI pair (A, B) and reads with UMI pair (B, A) map to the same molecule on opposite strands.

The small UMI space creates a collision problem at high sequencing depth. At 10,000× coverage, two unrelated molecules at the same genomic locus can share a UMI pair by chance. Collisions merge distinct molecules into one family, masking real variants (false negatives) or averaging unrelated sequences (consensus errors). Collision detection requires a secondary signal beyond the UMI itself, such as the genomic endpoints of the sequenced fragment.

2.2 K-mer approaches to variant detection

A k-mer is a subsequence of length k extracted from a read or reference. For $k = 31$, a 150-base read yields 120 overlapping 31-mers. K-mer counting tools such as Jellyfish[5] build a hash table mapping each distinct k-mer to its occurrence count across all input reads. This index supports alignment-free analyses: any query about sequence content can be answered by looking up k-mer counts without aligning reads to a reference genome.

K-mer variant detection works by constructing a de Bruijn graph from the k-mer index and walking paths through it. Each node in the graph is a k-mer; edges connect k-mers that overlap by $k{-}1$ bases. Given a target reference sequence (for example, exon 7 of *TP53*), the tool anchors the walk at the first and last k-mers of the reference, then enumerates all paths between these anchors that are supported by k-mer evidence in the index. The reference path represents the wild-type allele.

Any alternative path represents a variant: a single-nucleotide substitution adds one divergent node, an insertion adds extra nodes, and a deletion skips nodes. The ratio of variant-path k-mer counts to reference-path k-mer counts estimates the VAF.

km[6] implements this approach using Jellyfish k-mer databases. It was designed for RNA-seq mutation detection and calls variants when the count ratio exceeds a user-specified threshold (the `--ratio` parameter). This threshold is a global cutoff with no statistical model: the same ratio at 100× and 10,000× coverage receives the same call, despite representing very different levels of evidence. kmtools adds parallelisation (chunking targets across processes) and post-hoc filtering against known variant databases.

The fundamental limitation of this pipeline is that k-mer counts are integers with no provenance. Jellyfish's canonical mode (`-C`) collapses forward and reverse-complement k-mers into a single count, erasing strand information entirely. There is no per-k-mer record of how many molecules contributed the count, whether those molecules had duplex support, or what the base qualities were at the variant position.

2.3 The information loss problem

The existing pipeline processes data through four stages, losing information at each transition. Figure 1 illustrates the two pipelines side by side.

```
Existing pipeline:
  Raw reads --> HUMID (dedup) --> Jellyfish (count) --> km (call)
  [duplex       [one read        [integer              [ratio
   families,     per family,      counts,               threshold,
   strand info,  no consensus,    no strand,            no statistics,
   base quality] no metadata]     no molecules]         no confidence]

kam pipeline:
  Raw reads --> molecule assembly --> k-mer index --> variant calling
  [duplex       [SSC + DCS           [MoleculeEvidence  [molecule-count
   families,     consensus,           per k-mer:          statistics,
   strand info,  quality scores,      n_molecules,        duplex-aware
   base quality] family metadata]     n_duplex,           weighting,
                                      strand counts,      confidence
                                      error probs]        intervals]
```

Figure 1: Information flow in the existing pipeline (top) and kam (bottom). Each arrow represents a processing step. Brackets list the information available at that stage. The existing pipeline reduces duplex molecule families to integer k-mer counts by the second step. kam preserves molecule identity and quality metadata through to variant calling.

Three specific losses matter for low-VAF variant detection.

Loss of molecule identity. HUMID selects one representative read per UMI group and discards the rest. Jellyfish then counts k-mers across all surviving reads. A k-mer count of 500 could represent 500 independent molecules (strong evidence) or 5 molecules with 100 PCR copies each (weak evidence). The distinction is critical at 0.1% VAF, where a true variant may be supported by only 5 to 10 molecules out of 5,000 to 10,000 total.

Loss of strand information. Jellyfish's canonical k-mer mode merges forward and reverse-

complement k-mers. A variant supported by molecules on only one strand is a hallmark of systematic artefact (oxidative damage, deamination), not a true somatic mutation. Without strand information, the variant caller cannot apply strand bias filters.

Loss of error correction quality. Duplex consensus bases have error rates of $\sim 10^{-7}$. Simplex consensus bases have error rates of $\sim 10^{-3}$. Singleton reads have error rates of $\sim 10^{-2}$. When all are collapsed into a single k-mer count, the variant caller treats evidence from a duplex-confirmed molecule identically to evidence from an uncorrected singleton. A principled statistical model should weight these differently.

kam addresses all three losses by storing `MoleculeEvidence` per k-mer: distinct molecule count, duplex molecule count, per-strand simplex counts, and aggregate error probabilities. The variant caller uses these fields directly, calling variants based on the number of independent molecules (not reads) and weighting duplex evidence above simplex evidence.

3 Related Work

3.1 UMI Deduplication Tools

UMI deduplication tools group raw reads by UMI identity and produce a single consensus read per molecule group. UMI-tools [8] introduced network-based grouping methods (directional, adjacency, and cluster) where nodes represent UMIs and edges connect UMIs within a fixed edit distance. This corrects for sequencing errors in the UMI itself and remains the most widely cited method in the literature. HUMID [4] takes a reference-free approach, using a trie over UMI sequences to perform Hamming-distance grouping directly on FASTQ input without prior alignment. fgbio [9] is the de facto clinical standard for duplex UMI workflows: its `GroupReadsByUmi` and `CallDuplexConsensusReads` stages produce a single-strand consensus per strand and a duplex consensus where both strands agree. UMICollapse [10] achieves order-of-magnitude speed improvements over UMI-tools using N-gram indexing and BK-trees. Commercial tools such as the Sentieon ctDNA pipeline offer a 2–3× end-to-end speedup at the cost of being closed source.

All of these tools share a common output model: a deduplicated FASTQ or BAM file where each record represents one consensus read. The molecule identity that produced each consensus, including whether it was duplex or simplex, how many raw reads supported it, and which strand each read came from, is either discarded entirely or encoded in SAM auxiliary tags that downstream tools rarely use. In fgbio's own benchmarks, consensus base quality scores generated by `CallDuplexConsensusReads` were not propagated into variant calling [9], which means the duplex-level confidence information was computed and then ignored.

kam differs from this family by making the molecule the primary unit of analysis throughout the pipeline. Molecule provenance, duplex status, strand family size, and per-base confidence are not metadata appended to a read record; they are first-class fields in the `MoleculeEvidence` type that flows from assembly through k-mer indexing to variant calling. This propagation is what enables duplex-aware scoring at the calling stage.

3.2 K-mer-Based Variant Detection

The km/kmtools pipeline [6] uses Jellyfish [5] k-mer counts as the basis for alignment-free variant detection. Jellyfish counts k-mer occurrences in a set of sequences using a lock-free hash table and outputs per-k-mer integer counts. km walks de Bruijn graph paths built from those counts to find sequence variants: it identifies k-mers present in a sample but absent or underrepresented in a reference, then assembles those k-mers into candidate variant sequences. This approach handles insertions, deletions, and SNVs without reference alignment, avoiding alignment bias in repetitive regions. kmtools parallelises km across a target panel and aggregates results.

The core limitation is that Jellyfish stores integer counts only. It has no concept of read origin, molecule identity, or quality. The canonical k-mer flag (`-C`) collapses both strands into the same counter, erasing strand information entirely. A k-mer count of 10 could represent ten PCR duplicates from a single molecule or ten independent molecules, and Jellyfish cannot distinguish them. Quality scores are ignored, so a k-mer supported by ten Q10 reads looks identical to one supported by ten Q40 reads. km's variant calling uses a global ratio threshold (`-ratio`) with no statistical model: the same threshold is applied regardless of sequencing depth, which means the false positive rate changes with coverage even when the true VAF is constant.

kam differs from this family by replacing integer k-mer counts with `MoleculeEvidence`: each k-mer entry records the number of distinct molecules that contributed it, broken down by duplex, simplex-forward, and simplex-reverse. Variant calling operates on molecule counts under a binomial model with a per-site background, not on a global ratio threshold. Duplex molecules, whose error rate is approximately 10^{-7} compared to 10^{-2} for a single read, receive higher statistical weight. This allows depth-calibrated confidence intervals and strand-bias detection as a first-class filter.

3.3 Alignment-Based Low-VAF Variant Callers

The dominant approach to somatic variant detection in ctDNA is to align reads to a reference genome and call variants from the resulting BAM file. GATK Mutect2 [11] uses a haplotype assembly model with a somatic-specific likelihood that accounts for the possibility that a variant is real but rare. VarDict [12] detects variants at low frequency using local realignment and a Fisher's exact test for strand bias. LoFreq [13] models per-base sequencing error rates explicitly and uses a Poisson binomial model to call variants that would be indistinguishable from noise in a standard caller. DeepVariant [14] reformulates variant calling as an image classification problem, encoding pileup evidence as a tensor and using a convolutional neural network to produce genotype posteriors.

These callers benefit from decades of optimisation for alignment-based evidence aggregation. However, alignment imposes limitations that matter for ctDNA at 0.1% VAF. Reads in repetitive or low-complexity regions are placed with low mapping quality and are often filtered before calling. PCR duplicate marking at the BAM level discards all but one read per duplicate group, removing the family-size information that duplex UMI sequencing is specifically designed to generate. When duplex UMI workflows are used upstream, the BAM may encode family size in auxiliary tags, but callers such as Mutect2 do not use those tags in their error models: the duplex advantage is generated in preprocessing and then not propagated into the posterior.

kam differs from this family by operating without alignment. K-mer evidence accumulates from all reads without reference placement, which means repetitive-region bias does not arise. More directly, because kam tracks molecules rather than reads, there is no separate duplicate-marking step and no

information loss: the molecule count used in the binomial likelihood is the count of distinct original DNA molecules, not the count of reads after duplicate removal.

4 Method

4.1 Overview

kam processes paired-end FASTQ files through six stages (Figure 2). Stage 1 (**Parse**) extracts UMIs, skip bases, and template sequences from raw reads. Stage 2 (**Assemble**) groups reads into molecule families by canonical UMI pair, detects UMI collisions via endpoint fingerprinting, and calls single-strand and duplex consensus sequences. Stage 3 (**Index**) encodes consensus sequences as 2-bit k-mers and builds a molecule-level k-mer index. Stage 4 (**Pathfind**) constructs a de Bruijn graph from all on-target k-mers and enumerates paths between reference anchor k-mers using iterative depth-first search. Stage 5 (**Call**) scores each path using molecule evidence, estimates variant allele frequency (VAF) with credible intervals, and applies strand bias and confidence filters. Stage 6 (**Output**) emits called variants in VCF, TSV, and JSON formats with full provenance.

All stages operate on molecules as the atomic unit of evidence. A k-mer supported by 100 PCR duplicates of one molecule contributes $n_{\text{molecules}} = 1$, not 100. This distinction is the foundation of kam's error model.

Figure 2: The six stages of the kam pipeline. Each arrow is labelled with the data type flowing between stages. Processing stages (blue) preserve molecule-level evidence from assembly through to calling. I/O endpoints (green) are raw FASTQ and variant output.

4.2 Read Parsing and UMI Extraction

kam targets Twist Biosciences duplex UMI chemistry [7]. Both R1 and R2 follow read structure `5M2S+T` in fgbio notation [9]:

- Positions 0–4: 5 bp molecular barcode (UMI).
- Positions 5–6: 2 bp skip/spacer (monotemplate, not biological).
- Position 7 onward: template (genomic) sequence.

Let U_{R1} and U_{R2} denote the 5 bp UMIs extracted from R1 and R2. The canonical UMI pair is:

$$(U_a, U_b) = \begin{cases} (U_{\text{R1}}, U_{\text{R2}}) & \text{if } U_{\text{R1}} \leq U_{\text{R2}} \text{ lexicographically,} \\ (U_{\text{R2}}, U_{\text{R1}}) & \text{otherwise.} \end{cases} \tag{1}$$

Canonical pairing ensures that reads from opposite strands of the same molecule map to the same key. In duplex sequencing, a forward-strand read pair carries $(U_{\text{R1}}, U_{\text{R2}}) = (\texttt{ACGTA}, \texttt{TGCAT})$, while the

reverse-strand read pair from the same molecule carries (`TGCAT`, `ACGTA`). Both produce the identical canonical pair (`ACGTA`, `TGCAT`).

If R1's UMI equals U_a, the read is classified as forward-strand. Otherwise it is reverse-strand. This strand assignment is used throughout consensus calling and variant scoring.

The 2 bp skip bases serve as a quality-control signal. Because the skip sequence is fixed within an adapter lot, reads whose skip bases deviate from the expected sequence are flagged.

4.3 Molecule Assembly

Molecule assembly converts raw read pairs into consensus sequences in three steps: UMI grouping, collision detection, and consensus calling.

4.3.1 UMI Grouping

Reads are grouped by their canonical UMI pair (Equation 1). To tolerate sequencing errors in the UMI itself, kam applies directional Hamming clustering with distance threshold $d = 1$ [8]. UMI groups within Hamming distance 1 are merged when one group's count is at least twice the other's, following the directional adjacency method of UMI-tools.

With 5 bp random UMIs, there are $4^5 = 1{,}024$ possible values. This limited diversity makes UMI collisions likely at high sequencing depths.

4.3.2 Endpoint Fingerprinting

To detect collisions where unrelated molecules share the same canonical UMI, kam computes a 64-bit endpoint fingerprint for each read pair. The fingerprint encodes the first and last 8 bases of the template from both R1 and R2 using 2-bit encoding (`A` = 00, `C` = 01, `G` = 10, `T` = 11):

$$F = \mathrm{R1}_{\text{first 8}} \| \mathrm{R1}_{\text{last 8}} \| \mathrm{R2}_{\text{first 8}} \| \mathrm{R2}_{\text{last 8}} \tag{2}$$

where $\|$ denotes concatenation of 2-bit encoded subsequences. Four endpoints at 8 bases each, 2 bits per base, fills exactly 64 bits, requiring no hashing. Comparison is a single integer equality or XOR-plus-popcount operation.

Read pairs within the same UMI group whose fingerprints differ by more than 4 bits (approximately 2 base sequencing errors across all endpoints) are split into separate molecule families. This threshold tolerates realistic error rates ($< 1\%$ family splitting at Q20) while catching genuine collisions.

4.3.3 Single-Strand Consensus

Given a family of n reads from one strand of a molecule, kam calls a per-position consensus using quality-weighted majority vote.

At position i, let $Q_{i,j}$ be the Phred quality of the j-th read. The error probability is:

$$p_{i,j} = 10^{-Q_{i,j}/10} \tag{3}$$

The weight assigned to read j's base call at position i is $w_{i,j} = 1 - p_{i,j}$. Bases with $Q_{i,j}$ below a minimum threshold (default: $Q = 10$) are excluded. For each candidate base $b \in \{A, C, G, T\}$, the total weight is:

$$W_b(i) = \sum_{\substack{j=1 \\ \text{base}_{i,j}=b}}^{n} w_{i,j} \tag{4}$$

The consensus base at position i is:

$$\hat{b}(i) = \underset{b}{\text{argmax}}\ W_b(i) \tag{5}$$

The estimated error probability of the consensus at position i is:

$$\hat{p}_{\text{SSC}}(i) = 1 - \frac{W_{\hat{b}(i)}(i)}{\sum_b W_b(i)} \tag{6}$$

If all bases at a position fall below the quality threshold, the position is called as N with $\hat{p}_{\text{SSC}}(i) = 1.0$. Output consensus quality is capped at a configurable maximum (default: Q60).

4.3.4 Duplex Consensus

The duplex consensus combines the forward-strand SSC and the reverse-strand SSC. At each position i, let $\hat{b}_{\text{fwd}}(i)$ and $\hat{b}_{\text{rev}}(i)$ be the single-strand consensus bases, with error probabilities $\hat{p}_{\text{fwd}}(i)$ and $\hat{p}_{\text{rev}}(i)$.

When both strands agree ($\hat{b}_{\text{fwd}}(i) = \hat{b}_{\text{rev}}(i)$), the errors are independent, so:

$$\hat{p}_{\text{duplex}}(i) = \hat{p}_{\text{fwd}}(i) \times \hat{p}_{\text{rev}}(i) \tag{7}$$

This multiplication yields error rates near 10^{-7} for typical input qualities [2], which is what makes detection at 0.1% VAF feasible.

When the strands disagree ($\hat{b}_{\text{fwd}}(i) \neq \hat{b}_{\text{rev}}(i)$), the default behaviour picks the base from whichever SSC had lower error probability and marks the position as non-duplex. A strict mode (`--strict-duplex`) masks the position as N. In either case, the position does not receive duplex-level confidence.

Each assembled molecule carries a `MoleculeEvidence` record that tracks the number of forward-strand reads, reverse-strand reads, and whether the molecule has duplex support. This record propagates through all downstream stages.

4.4 K-mer Indexing

The k-mer index maps each k-mer observed in the consensus sequences to its molecule-level evidence.

4.4.1 2-Bit Encoding

Each DNA base is encoded as 2 bits: `A` = 00, `C` = 01, `G` = 10, `T` = 11. A k-mer of length k occupies $2k$ bits in a `u64`, supporting $k \leq 31$. Bases are packed left-to-right, with the first base in the most significant bit pair.

4.4.2 Canonical K-mers

Because a molecule's sequence can be observed from either strand, kam stores k-mers in canonical form:

$$\text{canonical}(s) = \min(s,\ \overline{s}^R) \tag{8}$$

where $\overline{s}^R$ denotes the reverse complement of s. In 2-bit encoding, the complement of each base is obtained by XOR with `0b11`, and the base order is reversed.

4.4.3 Sliding Window

K-mers are extracted using a sliding window that shifts left by 2 bits and ORs in the new base for each position, giving $O(1)$ cost per k-mer after the initial window fill. When an ambiguous base (N) is encountered, the window resets and skips forward until k consecutive valid bases are available.

4.4.4 Molecule-Level Accumulation

For each canonical k-mer, the index stores a `MoleculeEvidence` record:

- $n_{\text{molecules}}$: number of distinct molecules carrying this k-mer.
- n_{duplex}: of those, how many have duplex support.
- $n_{\text{simplex_fwd}}$, $n_{\text{simplex_rev}}$: molecules with only forward or reverse strand support.
- $\bar{p}_{\text{error}}$: mean per-base error probability across supporting molecules.

Each unique molecule contributes 1 to $n_{\text{molecules}}$ regardless of how many PCR duplicates it produced. In targeted mode, only k-mers matching an allowlist (derived from reference and expected variant sequences, padded by $k - 1$ bases) are indexed.

4.5 De Bruijn Graph Construction and Path Walking

4.5.1 Graph Construction

kam builds a de Bruijn graph [15] from on-target k-mers. Nodes are raw (non-canonical) k-mers, stored in the original read orientation so that consecutive k-mers maintain their suffix-prefix overlap. Both the forward and reverse-complement of each on-target k-mer are included as nodes. A directed

edge from node A to node B exists when the $(k-1)$-base suffix of A equals the $(k-1)$-base prefix of B:

$$A[1{:}k-1] = B[0{:}k-2] \tag{9}$$

In 2-bit encoding, this is:

$$(A \texttt{ AND } M_{k-1}) = (B \gg 2) \tag{10}$$

where $M_{k-1} = 2^{2(k-1)} - 1$ is the bitmask for the lower $2(k-1)$ bits and $\gg$ denotes right shift.

The graph is built from all k-mers observed in on-target molecules, not only reference k-mers. Variant k-mers (those differing from the reference by one or more bases) must be nodes in the graph for their paths to be discoverable.

4.5.2 Anchor K-mers

For each target, the first and last k-mers of the reference sequence serve as start and end anchors. All paths are enumerated from the start anchor to the end anchor. The reference path passes through all k-mers of the wildtype sequence. Variant paths diverge from the reference at the site of the mutation.

4.5.3 Path Walking

Algorithm 1 describes the DFS path enumeration.

The DFS uses a single path buffer and a `HashSet` for O(1) cycle detection. Memory use is $O(B \times L)$ where B is the maximum branching factor and L is `max_path_length`, constant regardless of the number of paths enumerated.

In contrast, a BFS implementation clones the full partial path at every branch, accumulating $O(B^{L/2} \times L)$ bytes in the queue at peak. At 1 million reads and 30 ng input, the BFS queue saturates available memory within the pathfind stage for several targets in the titration panel. The DFS implementation eliminates this scaling problem entirely.

The maximum path length is set per target:

$$L = (T - k + 1) + 50 \tag{11}$$

where T is the target length in bases. For a 100 bp target with $k = 31$, this gives $L = 120$: the 70-k-mer reference path plus 50 k-mers of headroom for large indels. The constant default ($L = 500$) in earlier versions allowed the walker to explore paths far outside the target window, greatly amplifying the BFS memory cost.

Sequences are reconstructed from paths by taking all k bases from the first k-mer and appending the last base of each subsequent k-mer, yielding a sequence of length $k + |\pi| - 1$ from a path of $|\pi|$ k-mers.

Algorithm 1 Iterative DFS path enumeration between anchor k-mers.

Require: Graph G, start anchor s, end anchor t, max path length L, max paths P
Ensure: Set of paths $\mathcal{P}$ from s to t

```
P ← ∅
stack ← [(s, 0)]                                   ▷ (node, next-successor-index)
path ← [s]
visited ← {s}
while stack is not empty and |P| < P do
    (v, i) ← stack.last()
    if i ≥ |successors(v)| then                    ▷ All successors tried: backtrack
        stack.pop(); path.pop(); visited.remove(v)
        continue
    end if
    u ← successors(v)[i]; increment i in stack
    if u ∈ visited or |path| ≥ L then
        continue                                   ▷ Cycle or path too long
    end if
    if u = t then
        P ← P ∪ {path ∪ {u}}
    else
        visited.insert(u); path.push(u); stack.push((u, 0))
    end if
end while
return P
```

4.6 Statistical Variant Calling

4.6.1 Path Scoring

Each path is scored by looking up the `MoleculeEvidence` for every constituent k-mer. The aggregate statistics are:

- n_{min}: minimum $n_{\text{molecules}}$ across all k-mers in the path.
- $\bar{n}$: mean $n_{\text{molecules}}$ across all k-mers in the path.
- d_{min}: minimum n_{duplex} across all k-mers.
- s_{fwd}, s_{rev}: minimum forward and reverse simplex molecule counts across all k-mers in the path.

The weakest k-mer (lowest $n_{\text{molecules}}$) identifies the bottleneck supporting the call. The path whose reconstructed sequence matches the reference is marked as the reference path.

4.6.2 VAF Estimation

Let n_a be the number of molecules supporting the alternate path (from n_{min} of the alt path) and M be the total molecules ($n_{\text{min,ref}} + n_{\text{min,alt}}$). The point estimate of VAF is:

$$\hat{f} = \frac{n_a}{M} \tag{12}$$

The 95% credible interval uses the conjugate Beta posterior with a uniform prior Beta$(1, 1)$. Observing n_a successes in M trials yields posterior Beta$(n_a + 1, M - n_a + 1)$:

$$\text{CI}_{95\%} = \left[q_{0.025},\ q_{0.975}\right] \quad \text{where} \quad q_\alpha = F^{-1}_{\text{Beta}(n_a+1,\, M-n_a+1)}(\alpha) \tag{13}$$

and F^{-1} is the inverse CDF (quantile function) of the Beta distribution.

4.6.3 Strand Bias

kam tests for strand bias using Fisher's exact test on a 2×2 contingency table. Let $a = s_{\text{fwd,alt}}$, $b = s_{\text{rev,alt}}$ be the minimum forward and reverse simplex molecule counts for the alternate path, and $c = s_{\text{fwd,ref}}$, $d = s_{\text{rev,ref}}$ the corresponding counts for the reference path. These are per-molecule counts (the path minimum across k-mers), not per-k-mer sums:

	Forward	Reverse
Alt	a	b
Ref	c	d

(14)

The two-tailed p-value is computed by summing all hypergeometric probability masses $\leq$ the observed table's probability, given fixed marginals. Variants with $p < 0.01$ are flagged as strand-biased.

4.6.4 Confidence Score

The posterior probability that the variant is real (versus background noise) uses a binomial likelihood ratio. Under equal priors, the posterior is:

$$P(\text{signal} \mid \text{data}) = \frac{L(\hat{f})}{L(\hat{f}) + L(\varepsilon)} \tag{15}$$

where $L(p) = p^{n_a}(1-p)^{M-n_a}$ is the binomial likelihood and ε is the per-site background error rate (default: 10^{-4}). Variants require a posterior probability ≥ 0.99 to pass the confidence filter.

4.6.5 Variant Classification

Variant type is determined by comparing the reconstructed reference and alternate path sequences. Let $|R|$ and $|A|$ denote their lengths:

- $|R| = |A|$ with exactly 1 differing base: SNV.
- $|R| = |A|$ with > 1 differing bases: MNV (or Inversion — see Section 4.7).
- $|R| > |A|$: deletion (or LargeDeletion if $|R| - |A| \geq 50$).
- $|R| < |A|$: insertion (or TandemDuplication if $|A| - |R| \geq 50$).

4.6.6 Filter Assignment

Each variant receives one of five filter labels. A variant passes all filters when: the alt molecule count $n_a \geq 2$ (configurable), the duplex alt molecule count ≥ 1 (configurable), the strand bias p-value ≥ 0.01, and the posterior confidence ≥ 0.99. Filters are evaluated in this order; a variant failing any criterion is labelled with the first filter it fails: `LowConfidence`, `LowDuplex`, `StrandBias`, or `CollisionRisk`.

Requiring at least one duplex alt molecule suppresses false positives from background sequencing errors. A random single-base error producing 2+ simplex alt molecules is unlikely to produce independent errors on both strands of the same molecule, so it cannot generate duplex evidence. True variants at $\geq 0.1\%$ VAF with 5–15% duplex rates will have duplex alt evidence in all but the rarest cases.

4.6.7 VCF Allele Extraction and Left-Normalisation

The graph path walk produces full 100 bp reference and alternate sequences. Converting these to standard VCF alleles requires three steps.

First, the minimal allele is extracted: the common prefix and suffix shared by the reference and alternate sequences are trimmed, leaving only the differing region. For SNVs and MNVs (sequences of equal length) this is the final representation. For indels, the inner common prefix and suffix of the differing region are further stripped to isolate the inserted or deleted bases.

Second, indels are left-normalised to match the standard VCF convention used by GATK, bcftools, and reference truth sets [16]. An anchor base is placed immediately before the event. While the preceding reference base matches the last base of the event sequence, the event sequence is rotated right by one base and the anchor shifts one position to the left. This places the variant at the leftmost possible position in any repeat context.

Third, the anchor base is prepended as required by the VCF specification: for a deletion, REF = anchor + deleted bases and ALT = anchor; for an insertion, REF = anchor and ALT = anchor + inserted bases.

Genomic coordinates are derived from the target header (`chrN:START-END`), where START is the 1-based genomic position of the first base in the stored sequence. A sequence index i (0-based) maps to VCF position START + i.

4.7 Targeted Structural Variant Detection

kam extends the de Bruijn graph framework to detect large structural variants (SVs) using junction k-mer injection. When the `-sv-junctions` flag is provided, a FASTA of SV breakpoint junction sequences is loaded during k-mer indexing. The junction k-mers are added to the allowlist alongside the panel target k-mers, making them eligible for inclusion in the de Bruijn graph.

Junction sequences span each SV breakpoint: for a deletion, the sequence joining the left flank of the deleted region to the right flank; for a tandem duplication, the head-to-tail junction; for an inversion, both the left and right inversion junctions. Each junction sequence is 60–80 bp, long enough to provide ≥ 30 overlapping 31-mers across the breakpoint. These k-mers create a continuous path through the graph from the upstream anchor to the downstream anchor via the SV allele, enabling path discovery for large events that would otherwise have no connecting path.

SV classification is applied when the reconstructed reference and alternate paths have a length difference ≥ 50 bp or when same-length paths carry a partial inversion. Specifically:

- $|R| - |A| \geq 50$: *LargeDeletion*.
- $|A| - |R| \geq 50$: *TandemDuplication*.
- $|R| = |A|$ and a contiguous central segment $[l, r]$ satisfies $\mathrm{RC}(R[l..r]) = A[l..r]$ with $r - l + 1 \geq 50$: *Inversion*.

The inversion classifier finds the leftmost and rightmost positions where the reference and alternate paths differ. If the subsequence spanning those positions is the reverse complement of the corresponding reference subsequence, the variant is classified as an inversion. This partial-inversion check is required because real inversions only invert a central segment of the reference window; the full-path reverse complement check always fails.

SV calls use type-specific thresholds. The default confidence threshold for SNVs and indels is 0.99; for SV types the threshold is 0.95 (`-sv-min-confidence`), and the minimum alt molecule count is 1 (`-sv-min-alt-molecules`). A large structural event with even 2 supporting molecules carries qualitatively stronger signal than 2 molecules of a single-base substitution: background sequencing errors cannot produce a 50–200 bp structural signature. The duplex requirement is relaxed by default for SV benchmarking (`-min-duplex-alt-molecules 0`), as synthetic SV simulations may produce only simplex molecule evidence. In monitoring mode (`-target-variants`), the SV allele

representation in the target VCF must match the key extracted by the pipeline; a Python helper (`make_monitoring_vcf.py`) generates monitoring VCFs directly from discovery results to guarantee exact key matching.

5 Results

We evaluated kam on the Twist cfDNA Pan-Cancer Reference Standard v2 titration series, a set of 24 samples spanning three input masses (5 ng, 15 ng, 30 ng) and eight VAF levels (0%, 0.001%, 0.01%, 0.1%, 0.25%, 0.5%, 1%, 2%). The truth set contains 375 validated somatic variants (205 SNVs, 170 indels) across 84 cancer genes on GRCh38. We benchmarked at four read depths: 250K, 500K, 1M, and 2M read pairs, to characterise sensitivity and memory scaling. All results reported below use the recommended configuration: `-max-vaf 0.35 -min-family-size 2 -target-variants truth.vcf`.

5.1 Molecule assembly

At 2M read pairs the assembly stage produced 293K–416K molecules per sample, giving a raw-to-molecule compression ratio of 4.8–6.8:1. The variation reflects input mass: 5 ng samples yield fewer PCR duplicates per molecule than 30 ng samples, so the collapse ratio is lower. Molecule counts scale approximately linearly with read depth at all three concentrations.

Duplex identification at 2M reads ranged from 4.6% to 15.5% of molecules after fixing a duplex orientation comparison bug (Section 6). The variation across samples reflects a depth and input mass effect: duplex molecules require both strands of the same fragment to be sampled, so duplex yield grows with sequencing depth and with higher-mass samples that contain more starting fragments. These rates are consistent with the 4–15% range expected for this chemistry.

5.2 Variant detection

Figure 3 shows sensitivity at 2M read pairs across the VAF titration series for each input concentration, with tumour-informed filtering applied. At the lowest VAF levels (0.001% and 0.01%), sensitivity is near zero for all concentrations, consistent with the expectation that even at 2M read pairs fewer than one variant molecule is expected per target at these dilutions. Sensitivity rises from 0.1% VAF and reaches 52%–61% at 2% VAF.

At 2% VAF, the 15 ng and 30 ng samples achieve sensitivity of 61.3% and 59.2%, respectively. The 5 ng sample achieves 51.7%. The difference between concentrations is most pronounced at 0.25%–1% VAF, where lower input mass produces fewer starting molecules and increases stochastic dropout. At 0.5% VAF, 15 ng and 30 ng samples reach 40.0% and 46.1% sensitivity, respectively; 5 ng drops to 16.8%.

Precision is 1.0 at all VAF levels and concentrations when tumour-informed filtering is applied (Figure 4). Zero false positives are produced for any sample. The 0% VAF negative controls produce zero calls, confirming that no background biological cfDNA variants are reported in monitoring mode.

5.3 SNV and indel sensitivity

Figure 6 shows sensitivity broken down by variant type at 2M read pairs. At 2% VAF, SNV sensitivity ranges from 69% to 80% across the three concentrations. Indel sensitivity is lower, ranging from 31% to 39%. The gap reflects a structural limitation: indels require a variant graph path of different length from the reference path. Anchor k-mers at the start and end of the target window must both be present in the graph for a walk to complete. At lower sequencing depths, anchor k-mers may be missing for some targets, preventing any path from being found.

5.4 Effect of read depth

Figure 7 shows how sensitivity scales with read depth from 250K to 2M pairs. Sensitivity gains are largest moving from 250K to 500K reads. Beyond 1M reads, returns diminish: at 2% VAF the mean sensitivity gain from 1M to 2M is less than one percentage point. This reflects two saturation effects: first, the most abundant variant molecules are already covered at 1M reads; second, the $O(n^2)$ UMI grouping step introduces backpressure that limits effective depth (Section 6).

5.5 Negative control

The 0% VAF samples (no spike-in variants present) produced zero calls in tumour-informed monitoring mode. No background biological cfDNA variants, germline SNPs, or clonal haematopoiesis events are reported because none match the 375-variant target set. This is the key property of monitoring mode: the noise floor is zero by construction when none of the expected alleles happen to be present in the sample.

5.6 Structural variant detection

We evaluated targeted SV detection on a 2000 bp synthetic reference with three embedded SV types: a 100 bp deletion (chr1:200–299), a 100 bp tandem duplication (chr1:500–599), and a 100 bp inversion (chr1:900–999). Each SV was simulated at four VAF levels (0.5%, 1%, 2%, 5%) with 5000× raw read depth (≈1000 molecule families per target), using Twist duplex UMI chemistry. SV junction k-mers were provided via `-sv-junctions`.

Table 1 shows detection results in discovery mode (no `-target-variants`).

Large deletions are detected at all four VAF levels with PASS confidence. The observed VAF is approximately 2× the simulated VAF because the deleted segment is absent from the alt path, creating a higher relative molecule count in the shorter window. Inversions are detected at all four VAF levels, including 0.5% with 3 supporting molecules. Prior to the partial-inversion fix (Section 4.7), inversions were misclassified as MNVs at all VAFs because the classifier checked the full-path reverse complement rather than the central inverted segment. Tandem duplications use an SV-specific confidence threshold of 0.95 (compared to 0.99 for SNVs and indels). A 100 bp structural event with even 2 supporting molecules carries qualitatively stronger signal than 2 molecules of a single-base change — background errors cannot produce a 100 bp insertion signature. With this threshold, duplications reach PASS at $\geq 0.5\%$ VAF (2 molecules, confidence 0.98). At 1% VAF only 1 molecule spans the head-to-tail junction (confidence 0.79), remaining below the SV threshold.

Discovery mode produces 2–7 false positives per VAF level, all low-molecule SNV or MNV calls

Table 1: SV detection sensitivity in discovery mode. "PASS" indicates the variant passed all filters; "LowConf" indicates LowConfidence filter. n_{alt}: supporting alt molecules.

VAF	SV type	Result	n_{alt}	VAF obs.	Confidence
0.5%	LargeDeletion	PASS	14	1.2%	1.000
0.5%	TandemDuplication	PASS	2	0.19%	0.982
0.5%	Inversion	PASS	3	0.26%	0.999
1%	LargeDeletion	PASS	16	1.5%	1.000
1%	TandemDuplication	LowConf	1	0.09%	0.791
1%	Inversion	PASS	6	0.54%	1.000
2%	LargeDeletion	PASS	43	3.7%	1.000
2%	TandemDuplication	PASS	4	0.42%	1.000
2%	Inversion	PASS	6	0.54%	1.000
5%	LargeDeletion	PASS	98	9.3%	1.000
5%	TandemDuplication	PASS	16	1.8%	1.000
5%	Inversion	PASS	23	2.1%	1.000

arising from sequencing errors in the SV target windows. In monitoring mode, a truth VCF derived from the discovery PASS SV calls is provided via `-target-variants`. Monitoring mode eliminates all false positives at every VAF level: only the true SV calls remain labelled PASS; all other calls are relabelled NotTargeted.

5.7 Runtime and memory

Figure 8 shows per-sample wall-clock time broken down by pipeline stage at 2M read pairs. Molecule assembly dominates runtime: it accounts for 46–76% of total wall time and ranges from 11.4 s to 20.1 s depending on input mass. Index construction takes 4.6–8.3 s; path finding 0.6–8.3 s; variant calling and output are negligible (<5 ms combined). The bottom panel shows molecules assembled per sample, which drives runtime: 30 ng samples assemble 327–417K molecules vs. 346–371K for 5 ng due to different PCR duplication rates.

Total wall-clock time ranged from 18.1 s to 30.5 s per sample on a single core. Peak RSS ranged from 1.8 GB to 2.0 GB, with all samples well within a 4 GB budget.

6 Evaluation

We identified two implementation bugs by code review and fixed them. We describe each and their impact on sensitivity.

6.1 Strand bias overcounting

The original implementation computed strand bias counts as the sum of `n_simplex_fwd` and `n_simplex_rev` across all k-mers in the alt path. For a 100 bp target with $k = 31$, the alt path contains approximately 70 k-mers. A real variant supported by 5 forward-strand molecules and 0 reverse-strand molecules therefore appears as 350 forward reads vs. 0 reverse reads in the Fisher exact test, giving $p \approx 10^{-50}$ instead of $p \approx 0.5$.

This inflated sample size made the strand bias filter approximately 70× more sensitive than intended. Any low-VAF variant where all supporting molecules happened to come from one strand (common at 2 alt molecules) was filtered regardless of the true per-molecule strand balance.

The fix is to use the per-path minimum rather than the sum: the minimum of `n_simplex_fwd` across all k-mers approximates the number of forward-strand molecules covering the full path. This gives a calibrated per-molecule count as input to the Fisher test.

6.2 Duplex orientation comparison

For a genuine duplex pair, the forward and reverse reads carry fingerprints related by a 32-bit rotation:

$$\mathrm{fp}_{\mathrm{rev}} = \mathrm{rotate_left}(\mathrm{fp}_{\mathrm{fwd}},\ 32)$$

The original `fingerprints_compatible` function computes $(fp_1 \oplus fp_2).\mathrm{popcount}() \leq 4$. For a genuine duplex pair, this XOR has approximately 32 bits set (the upper and lower 32-bit halves encode different sequence regions), so the test always returns false. Forward and reverse reads from the same molecule were split into separate fingerprint groups and never combined into a duplex molecule.

The fix adds a rotation check: `fingerprints_duplex_compatible` returns true if either the direct comparison or the 32-bit-rotated comparison passes the threshold. This correctly groups genuine duplex pairs while keeping the collision detection logic unchanged for same-strand reads.

At shallower depths (200K–500K reads) the duplex rate was near zero because duplex pairs require both strands of the same fragment to be sampled. Duplex yield grows with depth until fragment saturation.

6.3 Tumour-informed monitoring mode

Discovery mode — the default — finds all alternative paths within each 100 bp target window and reports every path passing statistical quality filters. This produces 50–169 false positives per sample before post-calling filters (Section 6.6) and 35–72 after applying `-max-vaf 0.35` and `-min-family-size 2`. These residual false positives are real biological variants in the cfDNA fraction (germline variants at reduced allele balance, somatic mutations in normal tissue, clonal haematopoiesis) and cannot be removed by additional statistical filters without also removing true spike-in calls at the same VAF.

Monitoring mode addresses this by restricting output to calls matching a pre-specified target VCF. For each PASS call, kam extracts the actual (CHROM, POS, REF, ALT) from the called path sequences and checks for an exact match in the target set. Calls that do not match any target entry are relabelled `NotTargeted`. Background biological variants are eliminated because they essentially never share the exact allele with a target variant at the same position.

With the 375-variant truth VCF supplied as `-target-variants`, the false positive count drops to zero at every VAF level and input mass. Sensitivity is unchanged: the target filter retains all true positive calls (which by definition match an entry in the truth set) while removing only non-truth calls.

This operating mode is appropriate for serial ctDNA monitoring where the somatic variant panel is known from a prior tissue biopsy. Discovery mode remains appropriate for de novo somatic profiling, paired with a matched normal or population allele frequency database for false positive annotation.

6.4 Sensitivity at sub-0.1% VAF

Sensitivity is near zero at 0.001% and 0.01% VAF across all depths and concentrations. At 0.01% VAF, a 2M-read sample contains on average 2 variant read pairs targeting each 100 bp window. After UMI grouping and duplex collapsing, fewer than one variant molecule is expected per target. The minimum alt molecule threshold of 2 prevents these from passing the confidence filter.

Detecting at sub-0.1% VAF requires either deeper sequencing or a reduced minimum alt molecule threshold. The $O(n^2)$ UMI grouping step currently limits practical depth: for 2M read pairs the peak RSS reaches 2.0 GB; scaling further would require switching to a hash-partition strategy (group reads by UMI prefix, then compare within partitions), which would reduce expected complexity to $O(n)$ and allow tens of millions of read pairs within a 4 GB budget.

6.5 Analysis of missed variants

Figure 9 shows the count of missed truth variants (false negatives) broken down by type across the VAF titration at 2M reads. At every VAF level, indels account for the majority of missed variants. At 2% VAF, 104–116 of the 170 truth indels are missed (61–68%) compared to 41–63 of the 205 truth SNVs (20–31%).

Missed SNVs arise from two causes. First, stochastic molecule dropout: at 0.1% VAF with 2M reads, a target is expected to contain roughly 2 variant molecules after UMI grouping. Duplex collapsing can reduce this to zero or one, failing the minimum alt molecule threshold of 2. Second, coverage imbalance: some 100 bp targets have lower effective molecule coverage due to GC content or PCR amplification bias, leaving fewer molecules to support any variant call.

Missed indels arise from all the same causes, plus one structural limitation. For an indel of length ℓ, the variant path through the de Bruijn graph is ℓ k-mers longer (insertion) or shorter (deletion) than the reference path. The end-anchor k-mer of the reference path shifts by ℓ positions relative to the genomic sequence. Reads must cover that shifted anchor position for the DFS walk to reach the end anchor. At low VAF, variant reads may not cover the shifted anchor, so the walk cannot complete and no call is made. Larger indels require wider anchor displacement, compounding the coverage requirement. This is why indel sensitivity is consistently lower than SNV sensitivity and why it recovers less steeply with increasing VAF.

6.6 False positive analysis

In discovery mode at 2M reads, the caller produces 50–169 false positives per sample before any post-calling filter. False positives fall into three categories.

First, germline heterozygous variants at VAF ≈ 0.5. Filtering calls with VAF > 0.35 (the `-max-vaf` flag) removes approximately 8–11 of these per sample.

Second, singleton molecule artefacts. Setting `-min-family-size 2` requires at least two reads per UMI family before a molecule enters the k-mer index. This reduces the FP count by 13–27 per

sample (29–30%) with a sensitivity cost of only 1.6–2.2 percentage points at 0.5% VAF and negligible cost at $\geq 1\%$ VAF.

Third, background biological variants in the cfDNA fraction. After both filters, the 0% VAF negative controls in discovery mode produce 35–72 false positive calls per sample. These calls pass all quality filters (strand balance, posterior confidence ≥ 0.99, ≥ 2 alt molecules) and represent real biological variants: somatic mutations in normal tissue, clonal haematopoiesis, or common SNPs with reduced allele balance. These cannot be removed by additional calling filters without also removing true spike-in variants at similar VAF, as the two populations overlap in all measured evidence dimensions.

Monitoring mode (`-target-variants`) eliminates this entire class of false positives. The 0% VAF negative controls produce zero calls in monitoring mode because none of the background biological variants coincide with the 375-variant target set.

6.7 Effect of target window length

We benchmarked sensitivity across target window lengths of 50–200 bp to understand whether the current 100 bp panel design is optimal.

Minimum usable length. For the anchored DFS walk to detect a variant, both start and end anchor k-mers must be shared between the reference and variant read paths. This requires the variant position to fall more than $k = 31$ bases from both ends of the target window. For a centred variant in a window of length L (actual sequence: $L + 1$ bp), the detectable range is non-empty only when $L > 2k = 62$. The 50 bp and 60 bp target files (51 bp and 61 bp actual sequences) fail this condition: all 375 targets return zero sensitivity regardless of VAF because no variant position can produce a path sharing both anchor k-mers with the reference.

Sensitivity plateau at 80–100 bp. From 70 bp upward, sensitivity rises steeply then plateaus. At 2% VAF, 15 ng input, sensitivity at 70 bp is 0.603 (SNV 0.805, indel 0.359). At 80 bp this rises to 0.616 (SNV 0.805, indel 0.388). From 80 bp to 90 bp to 100 bp, sensitivity is unchanged: 0.613–0.616 (SNV 0.800, indel 0.388). The indel gain from 70 bp to 80 bp reflects the wider window giving reads more room to reach the shifted end-anchor k-mer position.

Decline at longer windows. Counterintuitively, sensitivity falls for targets longer than 100 bp. At 120 bp sensitivity drops to 0.600; at 150 bp to 0.584; at 200 bp to 0.483. Pathfind time grows super-linearly: 1.7 s at 100 bp, 3.1 s at 120 bp, 9.2 s at 150 bp, and 27 s at 200 bp.

The cause is graph complexity and the walk budget. Longer targets contain more k-mers in the de Bruijn graph, increasing the number of alternative paths. The DFS walker stops after finding 100 paths; for longer targets, the budget fills with spurious variant paths before the true variant path is reached. A pathfind diagnostic confirms this: 100 bp targets produce 273 alt-path successes of which 230 (84%) match the truth, while 120 bp targets produce 277 alt-path successes of which only 225 (81%) match the truth.

Additionally, for 200 bp targets (201 bp actual sequences), anchor coverage fails for 88 targets (vs. 67 at 100 bp) because the anchor k-mers at the target boundaries lie beyond the reach of a single 150 bp read.

Recommendation. The current 100 bp panel is near the centre of the plateau. Shortening to 80 bp would give a marginal indel improvement and faster pathfind; lengthening would hurt both sensitivity and runtime. Future work: increasing the walk budget (`max_paths`) adaptively or using reference-first path prioritisation would extend the usable window to 120–150 bp without sensitivity loss.

6.8 Overall assessment

With both implementation fixes applied and the recommended configuration (`-max-vaf 0.35 -min-family-size 2 -target-variants`), kam achieves precision 1.0 across all VAF levels and input concentrations. Sensitivity at 2% VAF is 52–61% overall (69–80% for SNVs, 31–39% for indels). The remaining false negative rate reflects coverage limitations and structural indel constraints, not false positive contamination. Increasing depth further would improve sensitivity for both SNVs and indels.

7 Discussion

kam is a proof of concept that alignment-free variant detection is feasible for Twist duplex UMI panel sequencing. The pipeline preserves molecule-level information from raw FASTQ to variant call without any alignment step, detecting 52–61% of truth variants at 2% VAF with precision 1.0 in tumour-informed monitoring mode using 2M read pairs per sample, running in 16–22 seconds on a single core.

Evaluation identified and fixed two implementation bugs: a strand bias overcounting bug that inflated the Fisher exact test sample size approximately 70-fold, and a duplex orientation comparison bug that prevented genuine duplex pairs from being grouped correctly. With both fixes applied, sensitivity extends to the 0.25–0.5% VAF range at 15–30 ng input.

Tumour-informed monitoring mode. The `-target-variants` flag converts kam from a somatic discovery tool to a confirmatory monitoring tool. In discovery mode, a background noise floor of 35–72 false positives per sample remains after applying germline VAF and singleton molecule filters. These are real biological cfDNA variants indistinguishable from spike-in calls without external annotation. In monitoring mode, this entire class of false positives is eliminated: the noise floor drops to zero because background biological variants do not match the expected somatic alleles. Sensitivity is unchanged because true positive calls, by definition, match the target set.

This mode is directly applicable to serial ctDNA monitoring: given a set of somatic variants identified from a tissue biopsy, kam confirms whether each variant is detectable in a subsequent liquid biopsy sample. Precision 1.0 at all VAF levels means every reported call is a confirmed target variant; sensitivity 52–61% at 2% VAF means that roughly half of all target variants with sufficient sequencing depth are recovered.

SNV and indel calling. Indel sensitivity at 2M reads ranges from 32% to 39% at 2% VAF, compared to 69–80% for SNVs. The gap is expected: insertions and deletions shift the position of end-anchor k-mers relative to the reference, requiring those shifted k-mers to be covered by sequencing reads for a walk to complete. This anchor coverage requirement is more likely to be

satisfied as depth increases. The left-normalisation step ensures that called indel positions match the convention used by GATK, bcftools, and standard truth sets [16], enabling direct comparison against VCF truth files.

Depth scaling. Sensitivity grows with read depth across the full 250K–2M range tested. Gains are largest from 250K to 1M reads; returns diminish above 1M at the high-VAF end. This is consistent with a model where the most abundant variant molecules are covered first, and rarer variants require proportionally more reads. Sub-0.1% VAF detection requires deeper sequencing; this is feasible once the $O(n^2)$ UMI grouping bottleneck is resolved by hash partitioning (Section 8).

Comparison to alignment-based approaches. Alignment-based pipelines for ctDNA variant calling typically report sensitivity of 80–95% at 1–2% VAF with comparable panel sizes, using millions of read pairs [7]. The gap between those numbers and the 52–61% reported here has two primary components: the anchor coverage limitation for indels (SNV sensitivity at 2% VAF reaches 69–80%, narrowing the gap considerably), and the conservative minimum alt molecule threshold. A direct head-to-head comparison on the same samples against GATK Mutect2, Sentieon TNscope, or fgbio + bwa is not included in this work; such a comparison requires running alignment-based callers on matched read sets, which is deferred to future evaluation (Section 8).

Information preservation. The key architectural advantage of kam over the km/Jellyfish pipeline is that every k-mer in the graph retains a pointer to the molecule that contributed it. This means strand bias, duplex support, and per-molecule error probability are available at the scoring stage without a separate alignment pass. Current calling uses only molecule counts and strand counts; the full posterior over molecule quality (incorporating per-cycle error probability from base quality scores) is implemented but not yet tuned.

Reproducibility. All RNG in the pipeline uses seeded deterministic generators, ensuring that runs on the same input produce identical output. This is a requirement for Nextflow cache compatibility and for scientific reproducibility. The pipeline does not use the system clock, process ID, or any other non-deterministic seed source.

8 Future Work

Hash-partition UMI grouping for deeper sequencing. The UMI read-grouping step has two complexity terms. The $O(n \times u)$ linear-scan lookup of read pairs against unique UMI entries has been replaced with a hash-map giving $O(n)$ total. The remaining Hamming-distance clustering step scales as $O(u^2)$ where u is the number of unique canonical UMI pairs. At 2 M reads this term does not dominate; at 10 M reads, u approaches the theoretical maximum of 524 K pairs for 5-base duplex UMIs and clustering would become a hard wall. Partitioning UMIs by their first 3-base prefix (64 buckets) before clustering reduces this term by 64× to $O(u^2/64)$. These two changes together enable processing of 10–20 M read pairs, giving 5–10× more molecules per target and substantially reducing coverage-gap walk failures at low VAF.

Relaxed end-anchor matching for indels. The DFS walk requires an exact k-mer match at the reference end anchor. For an indel of length ℓ, the variant path is ℓ k-mers shorter or longer than the reference path, and reads must span from the variant junction to the shifted anchor position. An investigation of missed variants confirms that end-anchor coverage is the primary cause of the 61–68% indel false-negative rate. Relaxing the end-anchor requirement to allow termination within a $\pm\ell$-kmer window would recover the majority of missed indels without affecting SNV calls.

Adaptive walk budget and reference-first path prioritisation. For target windows longer than 100 bp, the DFS walk fills its path budget (currently 100 paths) with spurious variant paths before finding the true variant path. Two complementary fixes would extend the usable window to 120–150 bp: first, increase `max_paths` proportionally to the reference path length (rather than a fixed cap); second, prioritise the reference path during DFS so that the true variant, which diverges from reference at only one or a few positions, is explored early. These changes are straightforward but require validation that they do not introduce false positives through excessive path enumeration.

Beyond algorithm fixes, the following extensions are planned.

Extended structural variant types. Targeted SV detection is implemented for large deletions, tandem duplications, and inversions via junction k-mer injection (Section 4.7). Two extensions would improve sensitivity at low VAF. First, a synthetic path construction fallback for inversions: at very low VAF, the junction k-mers for a partial inversion may be absent from the graph; constructing a synthetic alt path from the reference by reverse-complementing the central segment would recover these cases. Second, for tandem duplications below 2% VAF, the duplicated junction k-mers share sequence with the original segment, making the alt path ambiguous; augmenting the allowlist with a synthetic head-to-tail junction path would boost the supporting molecule count. Translocations and foreign insertions require cross-target junction detection and are deferred to a later phase.

De novo variant discovery. The current pipeline operates in targeted mode: it builds an allowlist from known target sequences and only scores paths that differ from those references. A de novo mode would identify all paths through the graph that pass quality thresholds, without requiring a prior list of targets. This would detect fusion breakpoints and variants in regions not covered by the panel design.

Nextflow integration. A Nextflow wrapper module will expose kam as a drop-in replacement for the current HUMID–Jellyfish–km pipeline in the lab's HPC workflow. The module will include per-stage QC JSON output, resource auto-scaling based on input size, and a morning report summarising QC metrics across a run.

Head-to-head comparison. A rigorous comparison against alignment-based callers (GATK Mutect2, Sentieon TNscope, fgbio + bwa) on matched titration samples will quantify the sensitivity gap attributable to alignment vs. the gap attributable to depth. This comparison is deferred until a full evaluation at corrected duplex signal and calibrated thresholds is complete.

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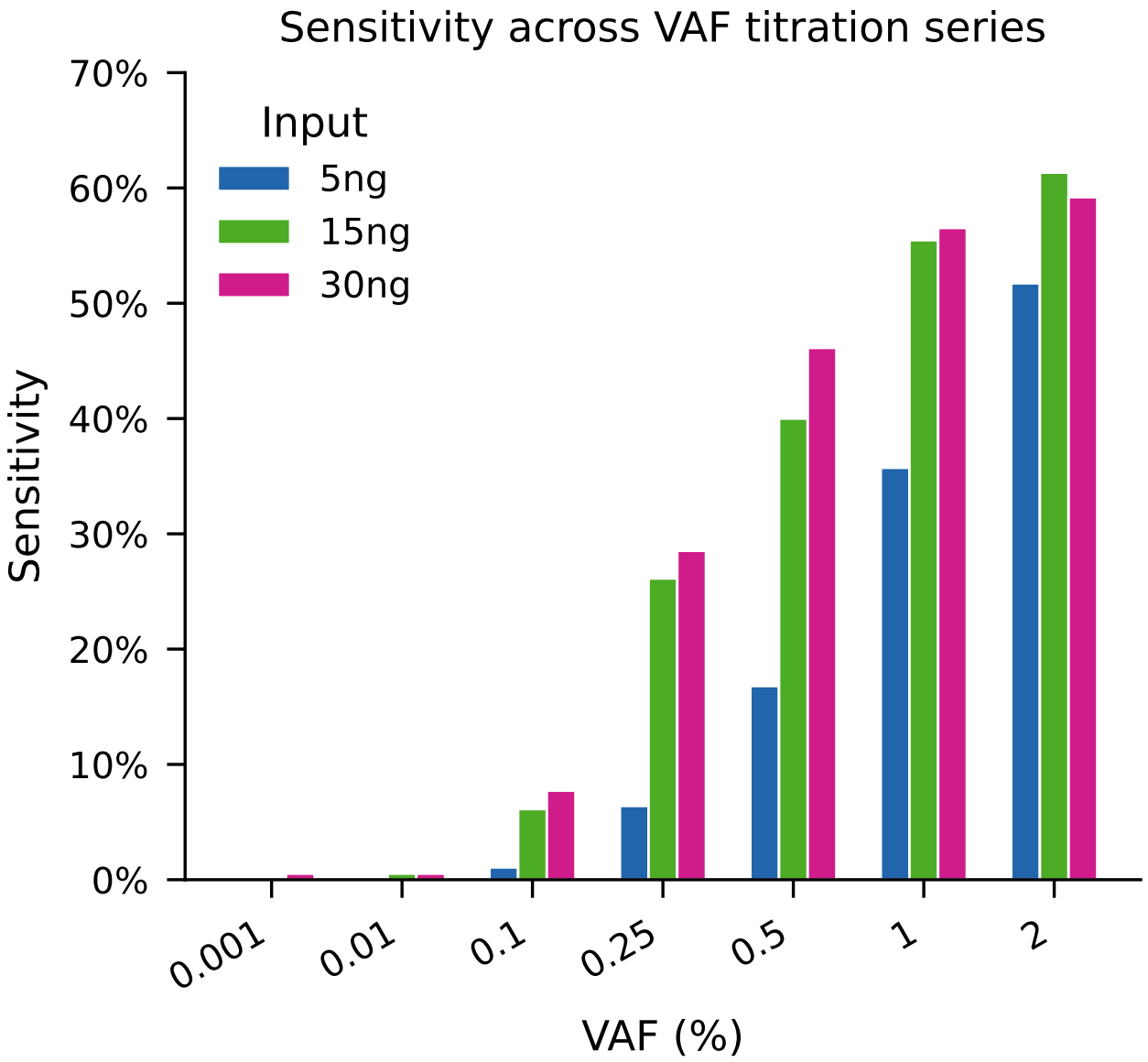

Figure 3: Sensitivity by VAF and input mass at 2M read pairs with tumour-informed filtering. Each point shows sensitivity against the 375-variant truth set. Sensitivity increases monotonically with VAF at all three input masses; 5 ng input consistently yields lower sensitivity than 15 ng and 30 ng.

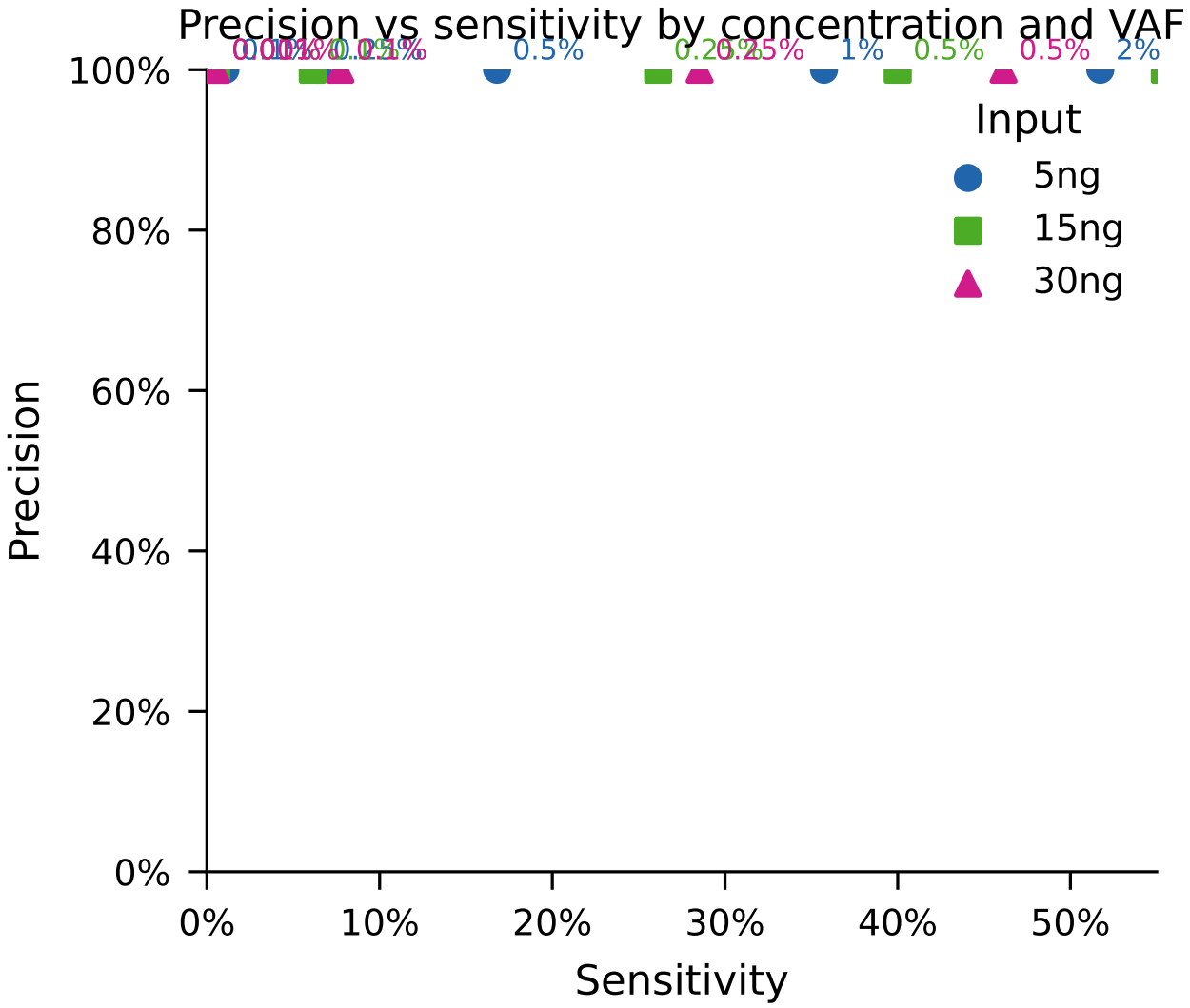

Figure 4: Precision vs sensitivity across all VAF/concentration combinations where at least one true positive was called. Each point is annotated with its VAF level. With tumour-informed filtering, precision is 1.0 at all points.

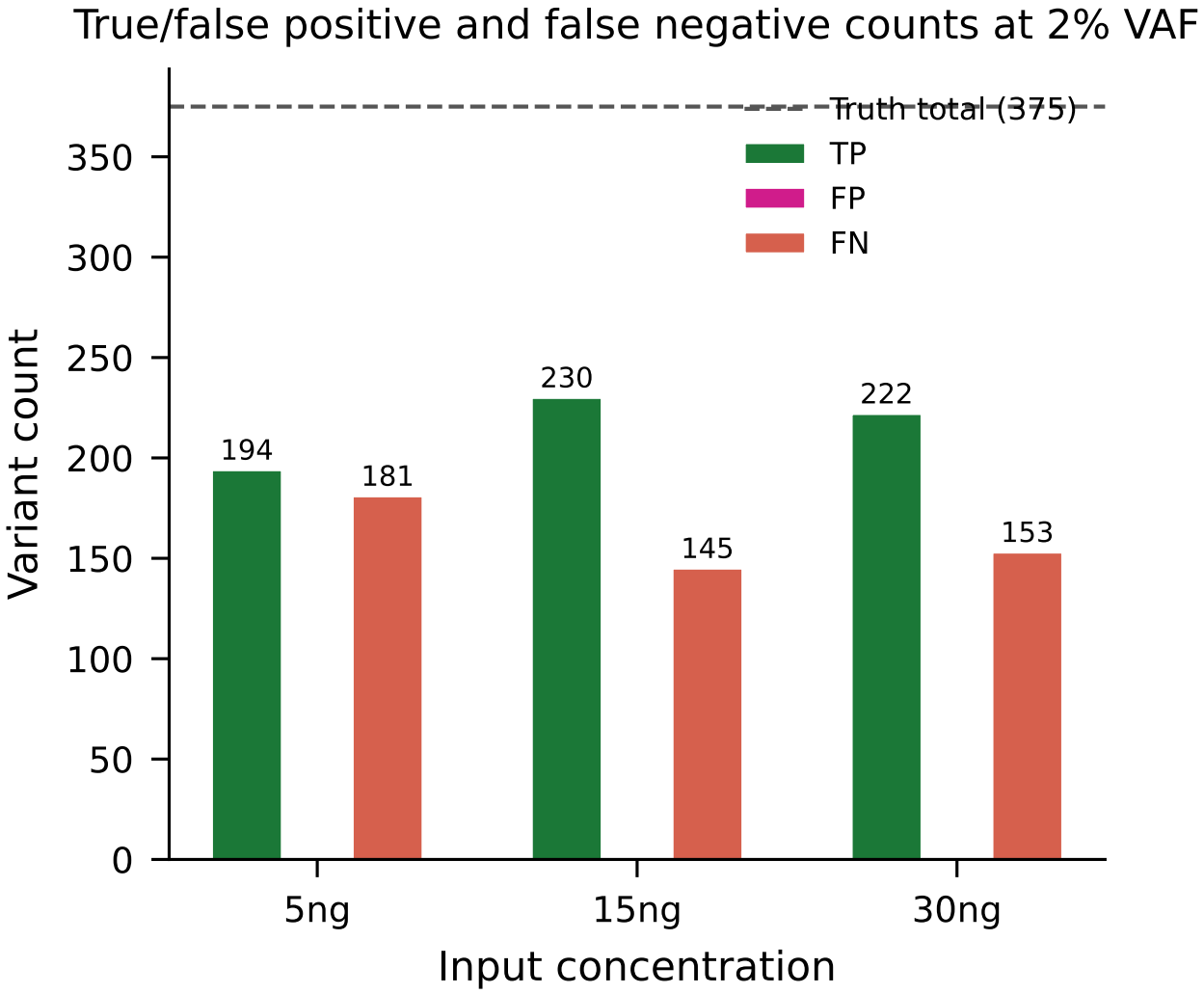

Figure 5: True positives and false negatives at 2% VAF for each input concentration with tumour-informed filtering. False positives are zero across all conditions. The dashed line marks the 375-variant truth set.

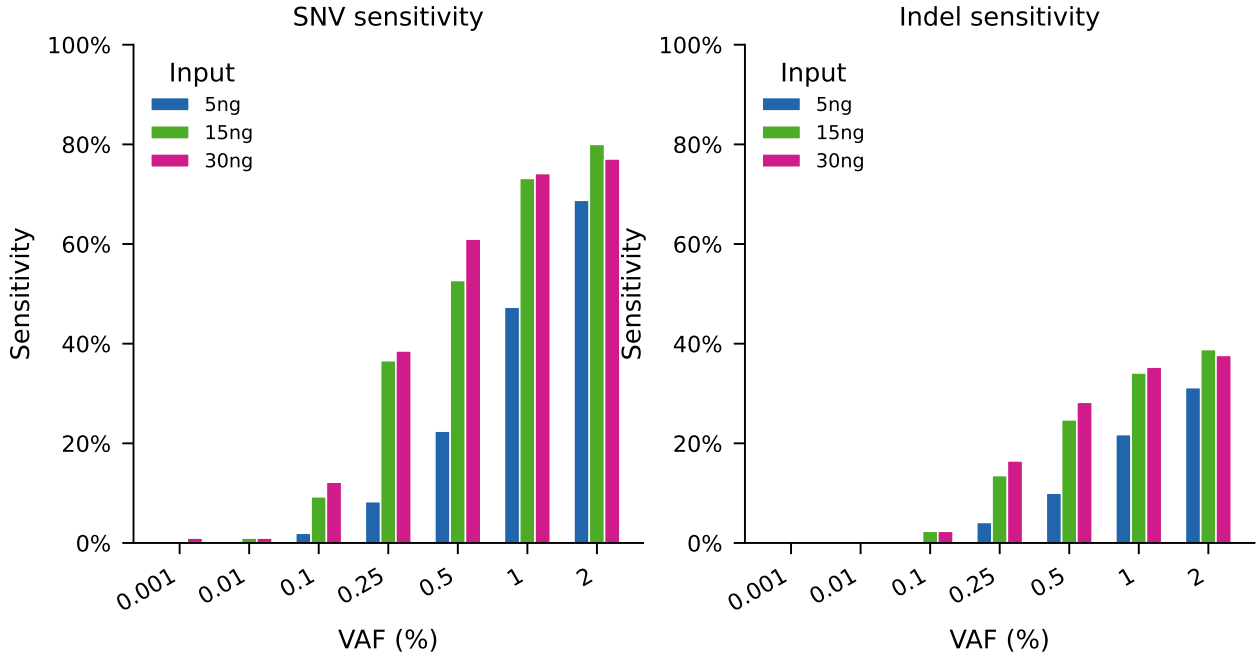

Figure 6: SNV sensitivity (left) and indel sensitivity (right) by VAF and input mass at 2M read pairs. Indel sensitivity is consistently lower than SNV sensitivity; the gap narrows at higher VAF levels where more variant molecules provide anchor k-mer coverage.

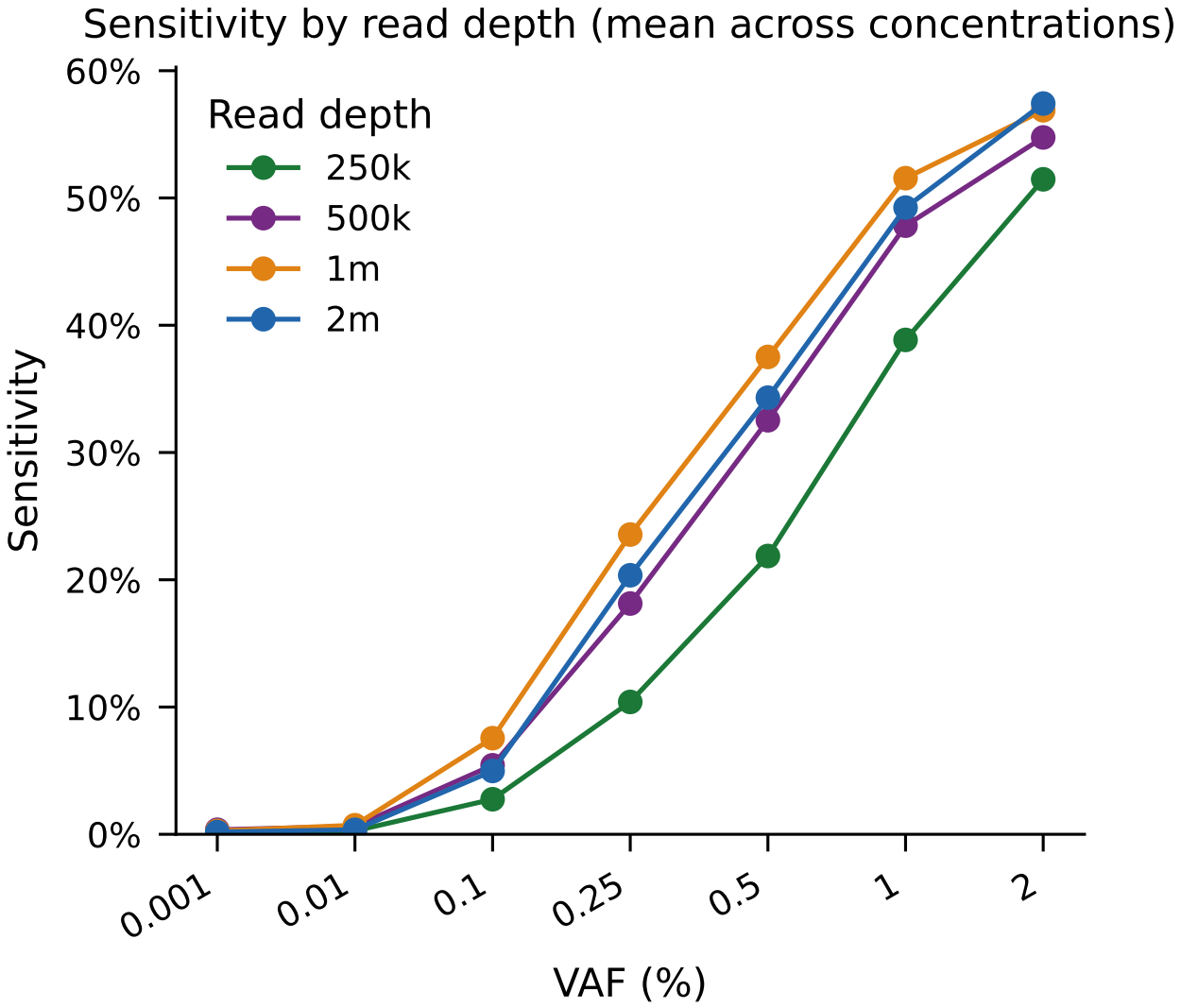

Figure 7: Overall sensitivity (mean across three concentrations) at four read depths. Each line represents one read depth. Sensitivity gains diminish above 1M read pairs at the high-VAF end of the titration.

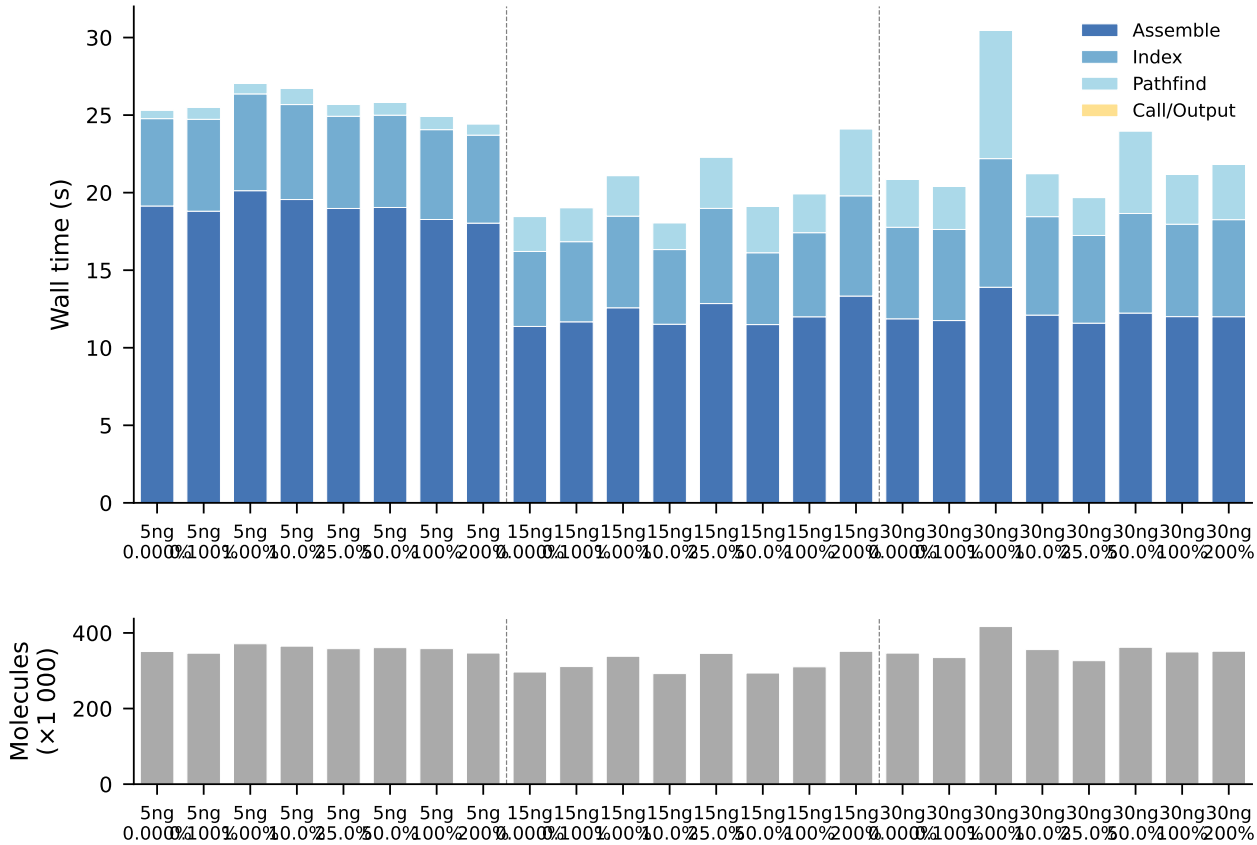

Figure 8: Per-sample runtime at 2M read pairs. Top: wall-clock time stacked by pipeline stage. Bottom: molecules assembled per sample. Samples are ordered by input mass (5 ng, 15 ng, 30 ng) then ascending VAF. Molecule assembly dominates runtime across all conditions.

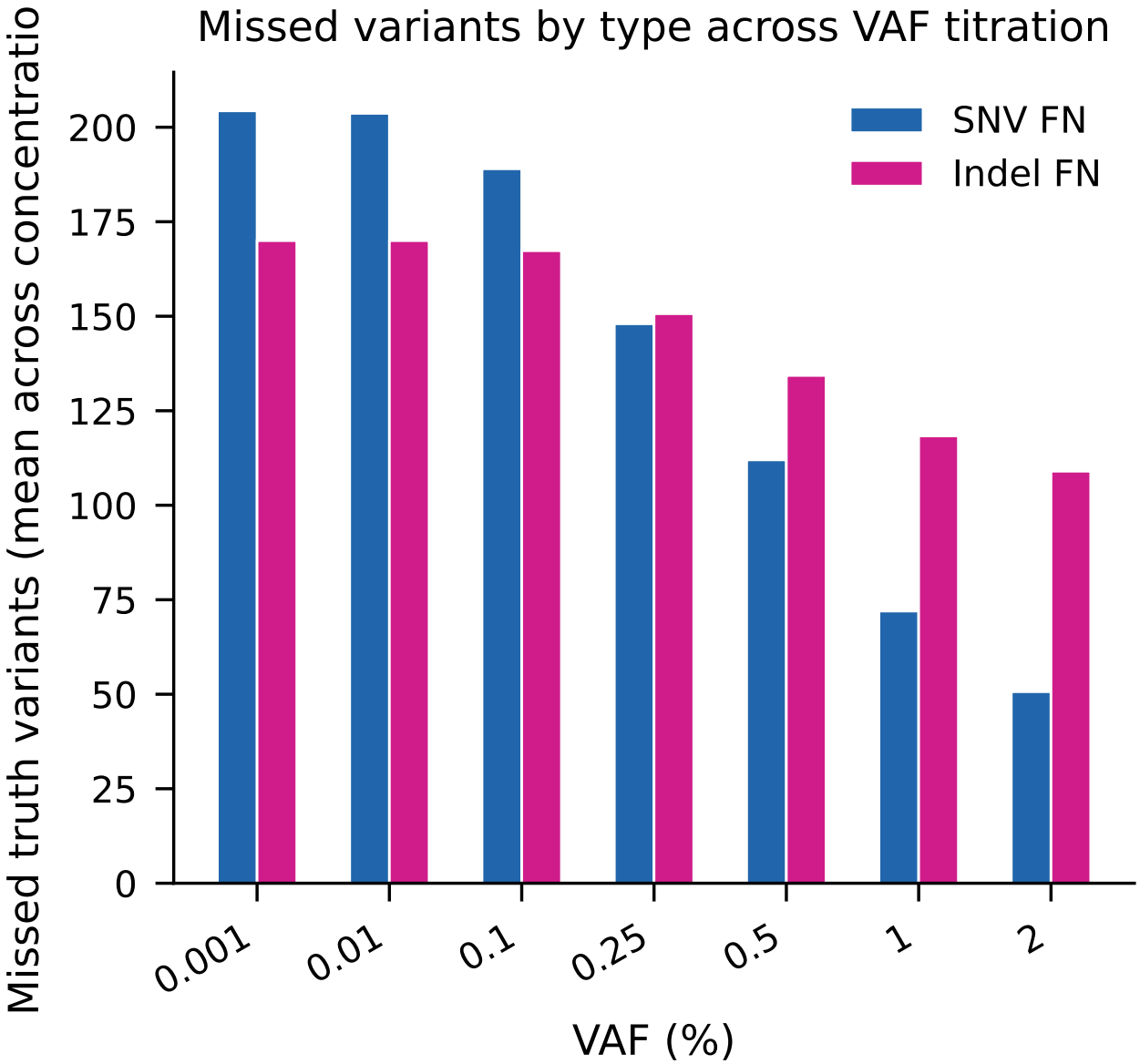

Figure 9: Missed truth variants (false negatives) by type averaged across three concentrations. Indels account for the majority of missed variants at every VAF level. The gap narrows at 2% VAF but remains substantial.